Mean Absolute Deviation

Lesson 8-3

Name: ______ Date: ______ Class: ______

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

Data: 8, 10, 12. Mean = 10. Distances from mean: 2, 0, 2. MAD = (2+0+2) ÷ 3 = 1.33

Mean Absolute Deviation

The average distance of each number from the mean.

If mean = 20 and value = 17, deviation = 17 − 20 = −3 (3 below the mean)

Deviation

How far a number is from the mean.

|−3| = 3 and |3| = 3 — both are 3 units from zero

Absolute Value

How far a number is from zero. It is always positive.

Low spread: 8, 9, 10, 11 (close together). High spread: 2, 9, 10, 25 (far apart)

Spread

How far apart the numbers are.

A set clustered tightly around the mean has low MAD; a set spread far from the mean has high MAD

Data distribution

How the data looks: where it sits and how spread out it is.

Low variability (MAD = 1): very consistent. High variability (MAD = 8): very spread out

Variability

How spread out the numbers are.

Key Ideas & Notes

- Coach wants to know which basketball player is more consistent: Player A scored 18, 22, 20, 24, 16 points in the last 5 games.
- Player B scored 10, 30, 25, 12, 23 points.
- Both players average 20 points per game.
- Which player would you count on to score close to 20 every night?

Think About It

- Both players have the same mean. What is different about their scores?
- Which player's scores stay closer to 20?
- Which player has the biggest single-game difference from 20?

My Notes

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Player A scored 18, 22, 20, 24, 16 and Player B scored 10, 30, 25, 12, 23. Both average 20. Which player would you count on to score close to 20 every night, and why?

Level 1 support

Start here: Two data sets can share a mean but differ in how spread out they are.

I would count on Player ____ because their scores are ____.

Contaría con el Jugador ____ porque sus puntajes están ____.

Even with the same mean, the players differ in ____.

Aunque tengan la misma media, los jugadores difieren en ____.

Word bank: mean absolute deviation | deviation | spread | consistent | mean

Listen for: *Students choose Player A as more consistent and recognize that equal means hide different spreads.*

Level 2

Predict: which player will have the larger MAD, and what does a larger MAD say about consistency?

- Player ____ will have the larger MAD because ____.
- A larger MAD means the scores are ____, so the player is ____.

EXPLORE

To find the MAD for Player A (18, 22, 20, 24, 16, mean 20), why do we take the ABSOLUTE VALUE of each deviation before averaging?

Level 1 support

Start here: Absolute value makes every distance from the mean positive so they don't cancel out.

We use absolute value so the deviations are all ___.

Usamos el valor absoluto para que las desviaciones sean todas ___.

Without absolute value, the deviations would add up to ___.

Sin valor absoluto, las desviaciones sumarían ___.

Word bank: **mean absolute deviation** **deviation** **absolute value** **spread** **mean**

Listen for: *A strong answer explains deviations above and below the mean would cancel to zero, so absolute value keeps each as a positive distance.*

Level 2

Compute and justify: find the MAD of 18, 22, 20, 24, 16. Show the deviations, take absolute values, and average them.

- The absolute deviations are ___, which sum to ___.
- Dividing by 5 gives a MAD of ___, meaning ___.

CONNECT

Two keepers both allow a mean of 1.67 goals, but Keeper A's MAD is 0.56 and Keeper B's is 1.22. Which keeper should the coach start in the championship, and why?

Level 1 support

Start here: A smaller MAD means more consistent, predictable performance.

The coach should pick Keeper ___ because their MAD is ___.

El entrenador debe elegir al Portero ___ porque su DAM es ___.

A smaller MAD means the keeper is more ___.

Una DAM más pequeña significa que el portero es más ___.

Word bank: **mean absolute deviation** **deviation** **spread** **consistent** **mean**

Listen for: *Students pick Keeper A and explain a smaller MAD (0.56) means goals allowed stay close to the mean, so performance is more reliable.*

Level 2

Critique: when might a coach actually prefer the LESS consistent keeper (higher MAD)? Defend with a scenario.

- A coach might prefer the higher-MAD keeper when ___.
- Higher spread could be acceptable if ___ because ___.

REFLECT

For the data set 4, 8, 6, 10, 2 with a mean of 6, talk through the steps to find the MAD.

Level 1 support

Start here: Find each deviation from the mean, take absolute values, then average them.

First I find each deviation from 6, then take the ___.

Primero hallo cada desviación de 6, luego tomo el ___.

I average the absolute deviations to get a MAD of ___.

Promedio las desviaciones absolutas para obtener una DAM de ___.

Word bank: **mean absolute deviation** **deviation** **absolute value** **spread** **average**

Listen for: *Students compute absolute deviations 2, 2, 0, 4, 4 (sum 12), divide by 5, and report MAD = 2.4.*

Level 2

Generalize: what does it mean about a data set if its MAD is exactly 0?

- A MAD of 0 means every value is ___.
- That happens because each deviation from the mean is ___.

Guided Examples

Example 1

The mean of a data set is 15. One value is 11. What is the absolute deviation of that value from the mean?

Solution: Deviation = 11 − 15 = −4. Absolute deviation = |−4| = 4.

Answer: A. 4

Example 2

A data set has absolute deviations of 3, 1, 5, 2, 4. What is the MAD?

Solution: Sum of absolute deviations = 3 + 1 + 5 + 2 + 4 = 15. MAD = 15 ÷ 5 = 3.

Answer: A. 3

Example 3

The mean of a data set is 20. A value is 26. What is the absolute deviation?

Solution: Deviation = 26 − 20 = 6. Absolute deviation = |6| = 6.

Answer: A. 6

Write About the Math

The Writing Revolution

I can explain my work using the words mean absolute deviation, deviation, absolute value, and spread.

1. Kernel Sentence

subject + verb

Model: Mean Absolute Deviation is the average distance of each number from the mean.
Desviación media absoluta es la distancia promedio de cada número a la media.

Write a kernel sentence about mean absolute deviation. Use a subject and a verb.
Escribe una oración base sobre desviación media absoluta. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Mean Absolute Deviation matters in math
Desviación media absoluta importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*	**Mean Absolute Deviation matters in math because ____.** *Desviación media absoluta importa en matemáticas porque ____.*
but *pero*	**Mean Absolute Deviation matters in math, but ____.** *Desviación media absoluta importa en matemáticas, pero ____.*
so *entonces*	**Mean Absolute Deviation matters in math, so ____.** *Desviación media absoluta importa en matemáticas, entonces ____.*

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about mean absolute deviation.
Di un hecho verdadero sobre mean absolute deviation.

Mean absolute deviation ____.

Question
Pregunta

Ask a question about mean absolute deviation.
Haz una pregunta sobre mean absolute deviation.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about mean absolute deviation.
Muestra entusiasmo sobre mean absolute deviation.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with mean absolute deviation.
Dile a un compañero qué hacer con mean absolute deviation.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

I would count on Player ____ because their scores are ____.
Contaría con el Jugador ____ porque sus puntajes están ____.

Even with the same mean, the players differ in ____.
Aunque tengan la misma media, los jugadores difieren en ____.

We use absolute value so the deviations are all ____.
Usamos el valor absoluto para que las desviaciones sean todas ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Team A has MAD = 2.1 points. Team B has MAD = 6.8 points. Which team is more consistent?

A. Team A — lower MAD means scores are closer to the mean

B. Team B — higher MAD means better performance

C. Both are equally consistent

D. Cannot tell from MAD alone

Show your work:

2. Two runners have the same average time of 60 seconds. Runner A's MAD is 1.2 seconds. Runner B's MAD is 5.8 seconds. Which runner is the coach more likely to pick for a relay race that needs a reliable time?

A. Runner A — lower MAD means more consistent times

B. Runner B — higher MAD means faster potential

C. Either — they have the same average

D. Neither — MAD doesn't matter for relay races

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Two basketball teams both average 75 points per game. Team A's last 5 scores: 73, 76, 74, 77, 75. Team B's last 5 scores: 60, 90, 65, 85, 75. Calculate the MAD for each team and explain which team a coach would prefer if they need predictable scoring.

Sentence starter: Team A's MAD = ____. Team B's MAD = ____. Team A is ____ because ____. A coach would prefer Team ____ for predictable scoring because ____.

Show your work:

Reflect — Exit Ticket

Data set: 4, 8, 6, 10, 2. The mean is 6. What is the MAD?

A. 2.4

B. 6

C. 0

D. 12

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. Team A — lower MAD means scores are closer to the mean *— Lower MAD means less spread — Team A's scores stay closer to their average, making them more consistent.*
2. **Try It 2:** A. Runner A — lower MAD means more consistent times *— Runner A's MAD of 1.2 means times are usually within 1.2 seconds of 60. Runner B's times vary more widely. For reliability, pick Runner A.*
3. **Exit Ticket:** A. 2.4 *— Deviations: −2, 2, 0, 4, −4. Absolute deviations: 2, 2, 0, 4, 4. Sum = 12. MAD = 12 ÷ 5 = 2.4.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Mean Absolute Deviation is the average distance of each number from the mean.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).